UNIVERSITE CATHOLIQUE D'AFRIQUE CENTRALE
FACULTES DES SCIENCES SOCIALES ET DE MANAGEMENT
Nkolbisson Yaoundé, B.P : 11628
Tel: (+237) 222 23 74 00
Fax: (+237) 222 23 74 02
Email: secretariat.fssg.nkolbisson@ucac-icy.net
Site: ucac-icy.net

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Continuous Assessment of Elements of Linear Algebra

Academic Year 2025-2026

(02 hours)

Exercise 1: (05 points)

Which of the following systems are linearly dependent? For those systems that are linearly dependent, find the nature of the dependence. For the rest, find their solutions.

(a)
$$\begin{aligned} 2x+\ 4y\ -z\ &=5\\ y\ +z\ &=2\\ x+\ y\ +z\ &=7 \end{aligned}$$

(b)
$$\begin{aligned} 4x-\ 2y\ +2z\ &=6\\ -y\ +z\ &=1\\ 2x\ &=2 \end{aligned}$$

(c)
$$\begin{aligned} -x_1+\ 2x_2\ -3x_3\ &=2\\ -2x_2\ &=3\\ 2x_1\ -x_2+x_3\ &=9 \end{aligned}$$

(d)
$$\begin{aligned} x_1+x_2+x_3+x_4\ &=1\\ -x_1+x_2\ -x_4\ &=1\\ x_1-x_2+x_3\ &=1\\ x_2+x_3\ &=1 \end{aligned}$$

(e)
$$\begin{aligned} 2x_1+\ x_3-2x_4\ &=5\\ -x_1+2x_2+x_3\ &=4\\ x_2-x_3+3x_4\ &=1\\ 2x_1-x_2+2x_3-5x_4\ &=4 \end{aligned}$$

Exercise 2:

I. Solve the linear system given explicitly. Show details **(05 points)**

1. $\begin{cases} 4x-6y=-11\\ -3x+8y=10 \end{cases}$
2. $\begin{cases} x+y-z=9\\ 8y+6z=-6\\ -2x+4y-6z=40 \end{cases}$
3. $\begin{cases} -2y-2z=-8\\ 3x+4y-5z=13 \end{cases}$

II. Showing the details, calculate the following expressions **(05 points)**